

Muhammad Adnan

PRINCIPAL SOFTWARE ENGINEER | AGENTIC AI | AZURE OPENAI | OPENAI API | .NET | ANGULAR | REACT
| GRAPHQL | MICROSERVICES | AZURE

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Summary

Principal Software Engineer with 11+ years of experience designing and delivering scalable enterprise applications using .NET, Angular, React, GraphQL, Microservices, and Azure Cloud technologies. Experienced in architecting distributed systems, optimizing application performance, and leading cross-functional engineering teams.

Hands-on experience building AI-powered enterprise solutions using OpenAI API and Azure OpenAI, including conversational AI, intelligent assistants, document processing, and business workflow automation. Skilled in integrating Large Language Models (LLMs) into existing enterprise systems and designing secure, scalable AI-enabled applications.

Passionate about modern software architecture, cloud-native development, and Agentic AI solutions that improve business productivity and operational efficiency.

Skills

- **Frontend Frameworks & Libraries (60%):** Angular, React.js, Next.js, Redux, RxJS, GraphQL, Material-UI, Tailwind CSS, Bootstrap
- **Backend Technologies (40%):** .NET Core, ASP.NET Web API, Entity Framework Core, C#, RESTful APIs, GraphQL APIs.
- **Performance Optimization:** Code Splitting, Lazy Loading, Virtual DOM, SSR, WebSockets, API Caching.
- **Cloud & DevOps:** Azure App Services, AWS, Docker, Kubernetes, Firebase, CI/CD Pipelines (Azure DevOps, GitHub Actions, Jenkins)
- **Databases:** SQL Server, PostgreSQL, MongoDB, Redis
- **Testing & Security:** Jest, Cypress, xUnit, SonarQube, OWASP Security Best Practices
- **Soft Skills:** Agile Development (SCRUM, Kanban), Performance Optimization, Code Reviews, Cross-Team Collaboration
- **Agentic AI & Generative AI:** OpenAI API, Azure OpenAI Service, Conversational AI, AI Assistants, LLM Integration, Prompt Engineering, Function Calling, Tool Calling, AI Workflow Automation, Retrieval-Augmented Generation (RAG), Semantic Search, Embeddings, Vector Databases, AI Orchestration, Context Management.
- **AI Frameworks & Platforms:** Azure AI Foundry, Semantic Kernel, OpenAI SDK, LangChain, LangGraph, AutoGen, CrewAI, Hugging Face Transformers.
- **Knowledge & Search Technologies:** Azure AI Search, Redis Vector Store, PostgreSQL pgvector, Pinecone, ChromaDB, Hybrid Search, Semantic Retrieval.

Experience

DMCR | **Principal Software Engineer** | **Agentic AI & Enterprise Solutions**

Mar-2025 - Present

- Architected and developed enterprise-grade **Agentic AI solutions** leveraging **OpenAI API** and **Azure OpenAI** services.
- **Designed and implemented AI Agents and Multi-Agent Systems** capable of orchestrating complex business workflows and autonomous task execution.
- **Built intelligent conversational AI** platforms and **enterprise chatbots** for customer support, knowledge retrieval, and business process automation.

- **Developed tool-calling and function-execution** capabilities enabling AI agents to interact with enterprise systems, APIs, and business services.
- **Integrated Large Language Models (LLMs)** into enterprise applications for document analysis, summarization, content generation, and intelligent search experiences.
- **Designed scalable AI microservices** using .NET Core and Azure cloud services, ensuring secure and high-performance AI workloads.
- Implemented prompt engineering and context management strategies to improve response **quality, reliability, and domain-specific accuracy**.
- Collaborated with business stakeholders to identify **AI automation** opportunities and translate business requirements into **production-ready** AI solutions.
- Led architecture decisions, code reviews, and technical mentoring activities across engineering teams while establishing best practices for AI-enabled applications.
- Worked with Azure cloud services and **CI/CD** pipelines to automate deployment and monitoring of **AI-powered** applications.

Unity Foods Ltd | Senior Software Developer (Team Lead)

Feb-2024 - Feb-2025

- **Refactored UI components in Angular**, ensuring modularity and maintainability.
- **Built backend services using .NET Core Web APIs with GraphQL**, improving API response times by **40%**.
- **Optimized SQL Server queries**, improving database response times by **30%**.
- **Developed dynamic, reusable React components** using best practices, reducing code duplication by **30%**.
- **Optimized GraphQL queries** and caching strategies, reducing API response times by **40%**.
- **Led migration from CSS frameworks to Tailwind CSS**, decreasing stylesheet size and improving maintainability.
- **Integrated WebSocket for real-time updates**, enhancing user interactivity and engagement.
- **Implemented server-side rendering (SSR) and static site generation (SSG)** to enhance SEO and performance.
- **Integrated GraphQL APIs** for efficient data fetching and reduced network overhead.
- **Implemented CI/CD pipelines with Azure DevOps**, automating deployments and reducing manual errors.

Progneo | Senior Software Developer (Team Lead)

Aug-2023 – Jan-2024

- **Built Angular-based admin panels** for internal applications, improving user efficiency.
- **Designed microservices using .NET Core**, reducing monolithic dependencies.
- **Implemented Redis caching for API responses**, reducing server load by **35%**.
- **Built large-scale React.js applications** with Next.js and Material-UI, improving performance and maintainability, leveraging SSR/ISR for enhanced SEO and performance.
- **Optimized Lighthouse performance scores** to achieve **95+** across multiple web projects.
- **Implemented GraphQL subscriptions** for real-time data fetching, reducing backend load by **35%**.
- **Enhanced accessibility compliance (WCAG 2.1)**, improving usability for a global audience.
- **Refactored legacy JavaScript codebases** into modern TypeScript projects, reducing runtime errors.
- **Managed Azure Cloud Services**, including Blob Storage, Key Vault, App Services, and CI/CD pipelines, ensuring secure and efficient deployments.
- **Created and managed Docker containers**, reducing deployment time by **30%**.
- **Utilized RESTful APIs** with token-based authentication, increasing system security and reliability.
- **Refactored RESTful API endpoints**, improving response times and security.

Dream Beyond Solutions | Senior Software Developer (Team Lead)

Oct-2022 – Aug-2023

- **Developed Angular web applications with .NET Core and ASP.NET Web API**, ensuring seamless frontend-backend integration.

- **Designed and prototyped applications using EF7 (model-first approach)** to modernize legacy MS Access applications.
- **Built microservices with SOA architecture, GraphQL, and Docker**, enabling scalable and containerized deployments.
- **Integrated Azure Services**, including Blob Storage, Key Vault, Functions, App Services, and CI/CD pipelines via Azure DevOps.
- **Developed customer-facing applications using ASP.NET MVC 5, CSHTML, JavaScript, Kendo UI, AJAX, and Bootstrap.**
- **Created RESTful APIs with authentication & authorization mechanisms using JWT and OAuth.**
- **Worked with SQL Server, MySQL, and PostgreSQL**, optimizing queries, stored procedures, and SSRS reports.
- **Built reusable UI components with Angular, Bootstrap, jQuery UI, and Kendo UI**, enhancing UI consistency.
- **Documented APIs using Swagger**, streamlining integration for frontend and external teams.

LITS Services | Senior Full Stack Developer

Jun-2022 – Oct-2022

- **Designed and developed Angular-based SPAs** using TypeScript and RxJS, improving reactivity and modularity.
- **Optimized Angular app bundle sizes**, reducing file load by 50% using lazy loading and module federation.
- **Integrated micro frontends architecture**, allowing multiple teams to work independently on different UI sections.
- **Developed backend services with .NET Core**, ensuring scalability and security.
- **Optimized Web API endpoints**, reducing API response times by **25%**.
- **Implemented authentication using JWT**, securing user access and data.
- **Developed UI dashboards** with high-performance charting libraries like D3.js and Chart.js.
- **Designed and developed SPAs using React**, improving UI reusability and accessibility.
- **Integrated third-party APIs**, enabling seamless external service interaction.

Catalyst IT Solutions | Software Developer

Feb-2015 – Jun-2022

- **Created frontend architectures for real-time trading platforms** using React, Redux, and WebSockets.
- **Developed stock market analytics dashboards**, reducing data processing latency by 30%.
- **Engineered a robust UI framework** that standardized components across multiple financial applications.
- **Built scalable, real-time trading dashboards** using React, Angular, and WebSockets.
- **Developed RESTful APIs in .NET Framework**, improving data transaction reliability.
- **Optimized SQL Server procedures**, reducing query execution time by **30%**.
- **Refactored legacy applications**, modernizing frontend and backend components.

Education

Quaid-e-Awam University of Engineering, Science & Technology

2017

BS Computer Science

Degree College, Nawabshah

2012

Intermediate

Certifications

- **Web Application Technologies and Django** - *University of Michigan* | 2021
- **Crash Course on Python** - *Google* | 2021
- **Python for Data Science, AI & Development** - *IBM* | 2021
- **Python Project for Data Engineering** - *IBM* | 2021

Projects

- **Enterprise AI Assistant using Azure OpenAI:** Designed AI-powered assistants using Azure OpenAI and OpenAI APIs to automate enterprise workflows, knowledge retrieval, and employee productivity tasks.
- **Multi-Agent Workflow Automation:** Developed AI Agent and Multi-Agent Systems capable of orchestrating business processes and autonomous task execution.
- **Intelligent Document Processing Platform:** Implemented LLM-powered document summarization, analysis, and information extraction capabilities.
- **API Gateway & Microservices Transformation:** Refactored monolithic enterprise applications into scalable microservices architectures.
- **E-commerce Platform Optimization:** Built a full-stack e-commerce site using **React, .NET Core, and SQL Server**, reducing checkout times by **40%**.
- **High-Frequency Trading Dashboard:** Built an Angular and React-based trading UI, handling **real-time data updates every 200ms**.
- **API Performance Enhancements:** Refactored GraphQL API consumption strategies, reducing request overhead by **40%**.
- **Enterprise Sales Dashboard:** Developed a **real-time data visualization tool** in React.js with Web API, improving insights for business teams.
- **API Gateway & Microservices:** Refactored a monolithic .NET application into microservices, improving maintainability and scalability.
- **Global Corporate Web Portal:** Led development of a **multi-region B2B enterprise web application** using Angular and .NET Core, improving customer engagement by **35%**.
- **Sales Portal for Corporate Customers:** Designed and implemented a sales portal for corporate customers, streamlining sales processes and improving customer engagement.
- **Stock Market Communication App:** Developed a real-time stock market communication app using C# and Windows Forms, enabling SMS and email notifications for trading activities.
- **B2B/B2C POS Solutions:** Designed and implemented POS solutions integrated with SAP, improving operational efficiency by 20%.

Languages

- **English: Fluent**
- **German: B1**